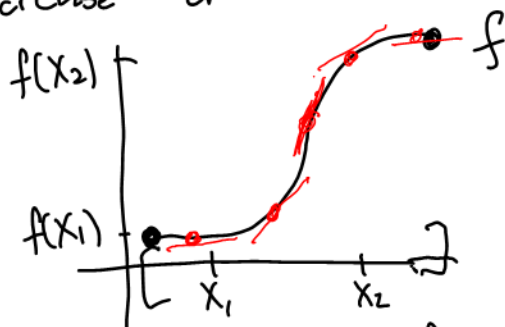


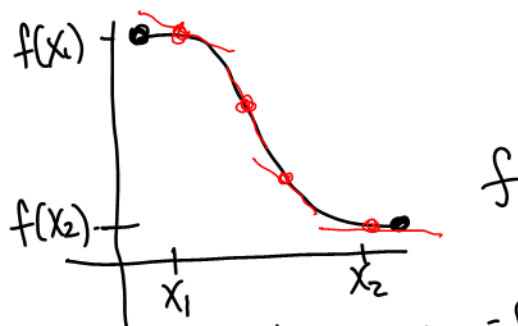
Lecture 17: ^{The} I/D and 1st Derivative Test

Goal: Use the 1st derivative to determine properties of the original function.

Recall what it means for a function f to increase or decrease on an interval.



f is increasing if
Whenever $x_1 < x_2$, $f(x_1) < f(x_2)$



f is decreasing if
Whenever $x_1 < x_2$, $f(x_1) > f(x_2)$

Theorem (I/D Test) Let f be a diff. fcn. on an open interval I

① If $f'(x) > 0$ for every x in I , f is inc on I

② If $f'(x) < 0$ for every x in I , f is dec. on I .

Example 1 When is $f(x) = \frac{x^2}{1+x^2}$ increasing and decreasing?

Step 1 Determine all critical numbers of f

$$f'(x) = \frac{2x(1+x^2) - (2x)x^2}{(x^2+1)^2} = \frac{2x + 2x^3 - 2x^3}{(x^2+1)^2}$$

$$= \frac{2x}{(x^2+1)^2} \quad \underline{\text{set } 0}$$

$$2x = 0$$

$$x = 0$$

Step 2 Use the critical numbers to divide the x -axis into regions to determine the sign of f' .

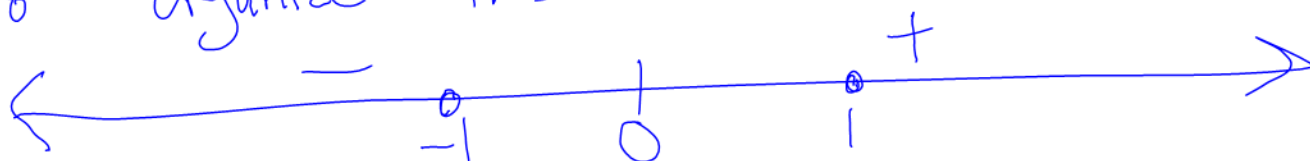
f'

	$\leftarrow$	-1	0	1	$\rightarrow$
Test Point:					
Sign of $2x$:		$2(-1) = -2$		$2(1) = 2$	
Sign of $(x^2+1)^2$		$+$		$+$	
Sign of f'		$-$		$+$	
Results of 1D Test		Dec	$\searrow \swarrow$	Inc	

Conclusion: f is increasing on $(0, \infty)$, while f is decreasing on $(-\infty, 0)$.

f has a rel. min at $x=0$.

NOTE: People often just use a number line to organize this information.



Example 2 let f be a polynomial such that

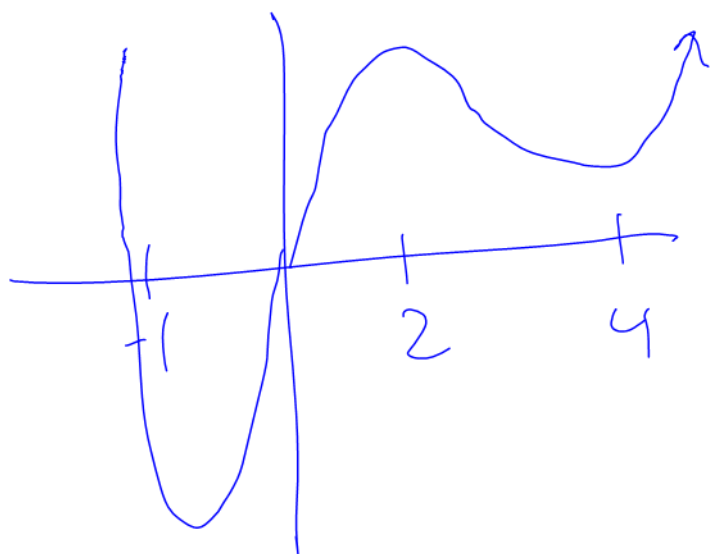
$$f'(x) = (x-2)(x+1)(x-4)$$

Determine when f is increasing and decreasing.

Step 1

$$(x-2)(x+1)(x-4) \stackrel{\text{set}}{=} 0$$

$$\Rightarrow x = 2, -1, 4 \Rightarrow x = -1, 2, 4$$



Step 2 Make "sign chart"

	$-\infty$	-1	0	2	3	4	∞
Test Point	-10^6						10^6
Sign of $x+1$	-		+		+		+
Sign of $x-2$	-		-		+		+
Sign of $x-4$	-		-		-		+
Sign of f'	-		+		-		+
Results of 1D Test	Dec	↘ ↗	Inc	↗ ↘	Dec	↘ ↗	Inc

Conclusion: f is increasing on $(-1, 2) \cup (4, \infty)$ and
 f is decreasing on $(-\infty, -1) \cup (2, 4)$
 f has rel. mins at $x = -1, 4$ and rel. max. at $x = 2$.

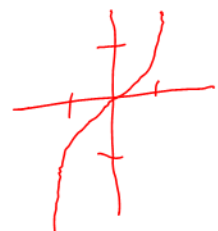
Example 3 Repeat for $f(x) = x^3$

Step 1 $f'(x) = 3x^2 \stackrel{!}{=} 0 \Rightarrow x = 0$

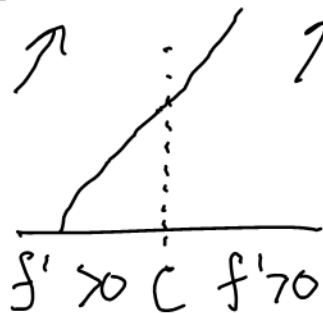
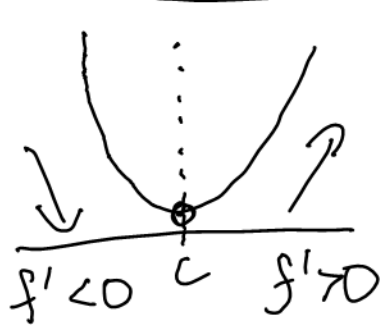
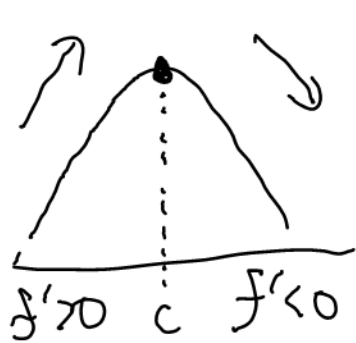
	$-\infty$	-1	0	1	∞
Test Point					
Sign of f'		+		+	
Results		Inc	↗ ↗	Inc	

Conclusion: f is increasing on $(-\infty, \infty)$ as opposed
to $(-\infty, 0) \cup (0, \infty)$

f has neither a rel. max nor min at 0.



The ID Test gives us a way to determine the locations of relative extrema (rel. maxs/mins)



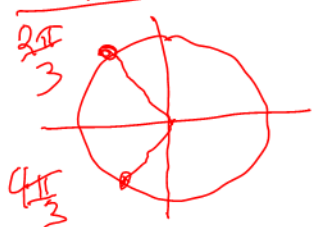
Theorem (1st Derivative Test) Let c be a crit. num. of a diff. fcn. f .

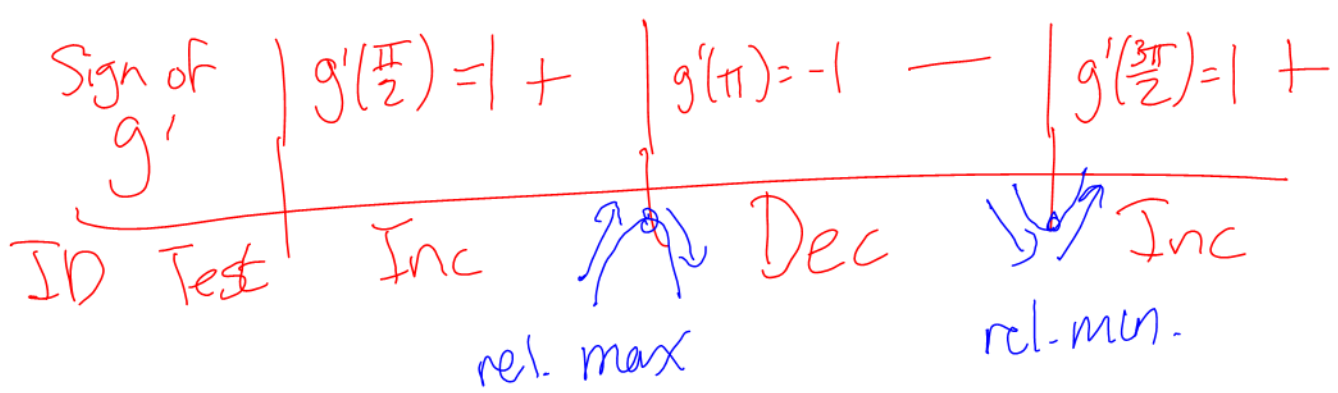
- ① If f' switches from $+$ to $-$ at c , then f has a rel. max. at c .
- ② If f' switches from $-$ to $+$ at c , then f has a rel. min. at c .
- ③ If f' does not switch signs at c , then f has neither a rel. max nor rel. min at c .

Example 4 The critical numbers of $g(x) = x + 2\sin x$ on $(0, 2\pi)$ are $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$. Determine the locations of any relative extrema on the interval $(0, 2\pi)$

Step 1 $g'(x) = 1 + 2\cos x$

Step 2





Ex5/Repeat for $f(x) = \frac{1}{8}e^{2x}(4x^3 - 6x^2 - 6x + 3)$

Step 1 Determine crit. nums.

$$f'(x) = \frac{1}{4}e^{2x}(4x^3 - 6x^2 - 6x + 3) + \frac{1}{8}e^{2x}(12x^2 - 12x - 6)$$

$$= e^{2x}(x^3 - \frac{3}{2}x^2 - \frac{3}{2}x + \frac{3}{4}) + e^{2x}(\frac{3}{2}x^2 - \frac{3}{2}x - \frac{3}{4})$$

$$= e^{2x}(x^3 - 3x)$$

$$= e^{2x}_x(x^2 - 3) \stackrel{!}{=} 0 \Rightarrow x = 0, \pm\sqrt{3}$$

Step 2 Make sign chart

	$\leftarrow \begin{array}{c} \quad \quad \quad \\ -\sqrt{3} \quad 0 \quad \sqrt{3} \\ \approx -1.7 \end{array} \rightarrow$			
Test Points	-100	-1	1	100
Sign of e^{2x}	+	+	+	+
Sign of $x^2 - 3$	+	$(-1)^2 - 3 = 1 - 3 = -2$ -	-	+
Sign of x	-	-	+	+
Sign of f'	-	+	-	+